Name of Student: Rikisha Bhattacharya	**PRN: 25070122155**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 9	
TITLE: Final Keyword in Java	

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

1. Create a Bank Account program where account number is final and cannot be changed.

Programme

Assignment9.java

src > Assignment9.java > Assignment9

```java
class BankAccount {

    final int accountNumber;
    String holderName;
    double balance;

    BankAccount(int accountNumber, String holderName, double balance) {
        this.accountNumber = accountNumber;
        this.holderName = holderName;
        this.balance = balance;
    }

    void display() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Holder Name: " + holderName);
        System.out.println("Balance: Rs. " + balance);
    }
}

public class Assignment9 {
    Run | Debug
    public static void main(String[] args) {

        BankAccount account =
            new BankAccount(accountNumber: 101, holderName: "Riya", balance: 50000);

        account.display();

        // account.accountNumber = 202;
        // This would produce an error because accountNumber is final.
    }
}
```

Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                    powershell
PS C:\Users\RIKISHA\PROGRAMMING\Java> cd src
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac Assignment9.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java Assignment9
Account Number: 101
Holder Name: Riya
Balance: Rs. 50000.0
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

2. **Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.**

Programme

Assignment9.java

```java
class Book {
    final String isbn;
    String title;
    String author;
    double price;

    Book(String isbn, String title, String author, double price) {
        this.isbn = isbn;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    void display() {
        System.out.println("ISBN: " + isbn);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: Rs. " + price);
    }
}

public class Assignment9 {
    Run | Debug
    public static void main(String[] args) {

        Book b = new Book(
            isbn: "9781234567890",
            title: "Java Programming",
            author: "James Gosling",
            price: 750
        );

        b.display();

        // b.isbn = "123456789";
        // Error: cannot assign a value to final variable.
    }
}
```

Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS          Debug: Assignment9
PS C:\Users\RIKISHA\PROGRAMMING\Java>  & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt
_socket,server=n,suspend=y,address=localhost:50868' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp
' 'C:\Users\RIKISHA\PROGRAMMING\Java\bin' 'Assignment9'
ISBN: 9781234567890
Title: Java Programming
Author: James Gosling
Price: Rs. 750.0
PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

Conclusion

This assignment helped us understand the use of the final keyword in Java. We learned how final can be used with variables, methods, and classes to prevent modification.

The program demonstrates how final provides security and consistency in Java programs.

GitHub Screenshot